Module 46

Transactions/1

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Week Recap

- Need for indexing database tables

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Module Objectives

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- To explore issues in concurrent execution of transactions

Module Outline

- Transaction Concept

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  - Concurrent execution of multiple transactions

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- The system should ensure that updates of a partially executed transaction are not reflected in the database

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  - ▷ Erroneous transaction logic can lead to inconsistency

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- However, executing multiple transactions concurrently has significant benefits

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- Transaction to transfer $50 from account A to account B:
- Once the user has been notified that the transaction has completed (that is, the transfer of the $50 has taken place), the updates to the database by the transaction must persist even if there are software or hardware failures

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  - This usually means that completed transactions (or their effects) are recorded in non-volatile

ACID Properties: Quick Reckoner

A - Atomicity

All or Nothing Transactions

C - Consistency

Guarantees Committed Transaction State

I - Isolation

Transactions are Independent

D – Durability

Committed Data is Never Lost

(c) http://blog.sqlauthority.com

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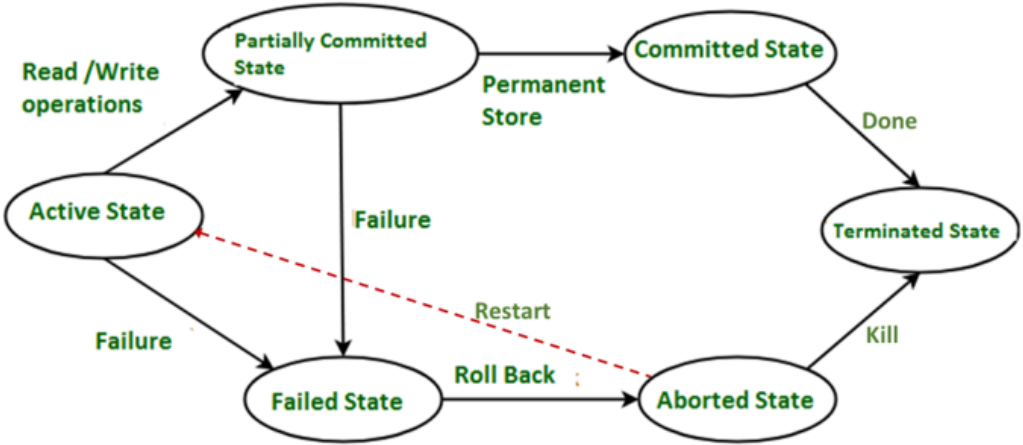
Transitions for Transaction States



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- To control the interaction among the concurrent transactions in order to prevent them from destroying the consistency of the database

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- Must preserve the order in which the instructions appear in each individual transaction
- A transaction that successfully completes its execution will have a `commit` instructions as the last statement
- By default transaction assumed to execute `commit` instruction as its last step

Schedules

- **Schedule**: A sequence of instructions that specify the chronological order in which instructions of concurrent transactions are executed
- A schedule for a set of transactions must consist of all instructions of those transactions
- Must preserve the order in which the instructions appear in each individual transaction
- A transaction that successfully completes its execution will have a `commit` instructions as the last statement
- By default transaction assumed to execute `commit` instruction as its last step
- A transaction that fails to successfully complete its execution will have an `abort` instruction as the last statement

Schedule 1

Example

- Let T1 transfer $50 from A to B, and T2 transfer 10% of the balance from A to B

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
	$B := B + temp$
	write (B)
	commit

Schedule 1

Example

- Let T1 transfer $50 from A to B, and T2 transfer 10% of the balance from A to B
- An example of a serial schedule in which T1 is followed by T2:

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
	$B := B + temp$
	write (B)
	commit

Schedule 2

Example

- A serial schedule in which T2 is followed by T1:

A	B	A+B	Transaction	Remarks
100	200	300	@ Start	
90	200	290	T2, write A	
90	210	300	T2, write B	@ Commit
40	210	250	T1, write A	
40	260	300	T1, write B	@Commit

Consistent @ Commit
Inconsistent @ Transit
Inconsistent @ Commit

Values of A & B are different from Schedule 1 - yet consistent

Schedule 3

Example

- Let T1 and T2 be the transactions defined previously. The following schedule is not a serial schedule, but it is equivalent to Schedule 1

Note - In schedules 1, 2 and 3, the sum 'A + B' is preserved

Schedule 4

Example

- The following concurrent schedule does not preserve the sum of 'A + B'

Module Summary

- A task in a database is done as a transaction that passes through several states

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Module Summary

- A task in a database is done as a transaction that passes through several states
- Transactions are executed in concurrent fashion for better throughput

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- A task in a database is done as a transaction that passes through several states
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- Concurrent execution of transactions raise serializability issues that need to be addressed

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Module Summary

- A task in a database is done as a transaction that passes through several states
- Transactions are executed in concurrent fashion for better throughput
- Concurrent execution of transactions raise serializability issues that need to be addressed
- All schedules may not satisfy ACID properties

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